Quiz — 1.3.4 Web Technologies

Name: ____ Date: _ ***Mark:*** ___ / 25

OCR H446 · Component 01 · 1.3.4

Section A — Multiple choice (8 marks)

1. The role of HTML in a web page is to define the: A. Colour and font styling B. Structure and content C. Server-side database logic D. Encryption of the page

Your answer: ☐

2. Which technology is primarily responsible for the presentation/styling of a page? A. HTML B. CSS C. JavaScript D. SQL

Your answer: ☐

3. A CSS selector beginning with `#` targets an element by its: A. tag name B. class C. id D. attribute value

Your answer: ☐

4. JavaScript is most commonly used to provide: A. Page structure B. Client-side interactivity and behaviour C. Database storage D. IP addressing

Your answer: ☐

5. In PageRank, a link from one page to another is treated as: A. A penalty B. A "vote" for the linked page's importance C. A keyword in the index D. An encryption key

Your answer: ☐

6. Links from high-ranking pages in PageRank are: A. Ignored B. Worth more than links from low-ranking pages C. Always worth the same as any other link D. Counted as negative votes

Your answer: ☐

7. Which is an example of server-side processing? A. Instant form validation in the browser B. Animating a button on hover C. Checking login credentials against a database D. Showing/hiding a menu on click

Your answer: ☐

8. Why must security-critical validation also be performed server-side? A. It runs faster on the client B. Client-side checks can be viewed and bypassed by the user C. The server cannot run JavaScript D. It reduces server load

Your answer: ☐

Section B — Short answer (12 marks)

9. State the role of each of HTML, CSS and JavaScript in a web page. **[3]**

10. Describe how a search engine builds its index of web pages. **[2]**

11. Explain the purpose of the damping factor in the PageRank algorithm. **[2]**

12. PageRank is calculated iteratively. Explain what this means and when it stops. **[2]**

13. Give one advantage and one disadvantage of using client-side processing instead of server-side processing. **[2]**

14. State one task that is better suited to server-side processing and explain why. **[1]**

Section C — Exam-style question (5 marks)

15. An online shop has a checkout form. When a customer submits the form, the email address format is checked instantly in the browser, and the payment details are then verified against the bank's records.

Explain why the email check can be done client-side but the payment verification must be done server-side, referring to speed, server load and security. **[5]**

Answer key

Section A (8 marks, 1 each)

1. **B** — HTML defines structure and content.
2. **B** — CSS handles presentation/styling.
3. **C** — `#` selects by id (`.` selects by class, a bare name selects by tag).
4. **B** — JavaScript provides client-side interactivity/behaviour.
5. **B** — Each incoming link counts as a vote for importance.
6. **B** — Votes from high-rank pages carry more weight.
7. **C** — Verifying login/credentials against a database is server-side.
8. **B** — Client-side checks can be viewed/altered/bypassed, so are not secure on their own.

Section B (12 marks)

9. **[3] (1 mark each)** HTML: structure and content (headings, paragraphs, links, images). CSS: presentation/style (colour, font, layout). JavaScript: interactivity/behaviour, usually client-side (events, validation, dynamic updates).

10. **[2]** 1 mark: a web crawler/spider follows links and reads/visits pages. 1 mark: the engine builds an index mapping keywords to the pages they appear on for fast retrieval (and ranks by relevance).
11. **[2]** 1 mark: the damping factor (~0.85) models the probability a "random surfer" keeps following links rather than jumping to a random page. 1 mark: it prevents pages from gaining unrealistically high/looping rank and ensures the calculation converges to sensible values.
12. **[2]** 1 mark: the rank of each page is recalculated repeatedly using the latest rank values of the pages linking to it. 1 mark: it stops when the values stop changing significantly (converge).
13. **[2]** 1 mark advantage: faster/immediate response and reduced server load. 1 mark disadvantage: insecure (can be viewed/bypassed) / depends on the user's browser/device capabilities.
14. **[1]** Any valid: e.g. database queries, login authentication, or payment checks — because the data/logic must be kept secure and hidden, and cannot be trusted to the client.

Section C (5 marks — award up to 5 from)

15.
    - The email format check is simple and not security-critical, so doing it client-side gives the user instant feedback. **[1]**
    - Doing it client-side reduces load on the server because no round trip is needed for a simple check. **[1]**
    - Payment verification must be done server-side because it is security-critical — client-side code can be viewed and bypassed by the user. **[1]**
    - The bank's records/logic must stay hidden on the server, so the verification cannot be trusted to the browser. **[1]**
    - Server-side processing is secure but slower (a round trip to the server) and adds server load — an acceptable trade-off for sensitive payment data. **[1]**

Total: 8 + 12 + 5 = 25 marks